

Introduction to Computer and Computer Science

Homework Assignment #9

Due: 12/2 0:00 am

In this homework assignment, you can use any concept you have learned from the lecture to complete the task.

Binary & Hexadecimal Representations

Binary and hexadecimal representations are very common in the realm of computer science. In this homework, we are going to develop a program, which aims to transform an arbitrary decimal number into either a binary number or a hexadecimal number.

Problem #1: Binary representation

Please write a function called “**getBinaryRep**” which can transform an arbitrary decimal number into a binary number. It should have two parameters as shown below,

getBinaryRep(n, numDigits),

where n is the target decimal number and numDigits represents how many digits you are going to use in the binary representation.

Problem #2: Hexadecimal representation

Please write a function called “**getHexRep**” which can transform an arbitrary decimal number into a hexadecimal number. It should have two parameters as shown below,

getHexRep(n, numDigits),

where n is the target decimal number and numDigits represents how many digits you are going to use in the hexadecimal representation.

You can test your functions by these cases:

Case 1

Operation: `print(getBinaryRep(15, 4))`

And the program should return you: 1111

Case 2

Operation: `print(getBinaryRep(58760, 16))`

And the program should return you: 1110010110001000

Case 3

Operation: `print(getHexRep(225, 4))`

And the program should return you: 00E1

Case 4

Operation: `print(getHexRep(9527, 4))`

And the program should return you: 2537

Case 5

Operation: `print(getBinaryRep(16, 4))`

And the program should return you: ValueError: not enough digits

Case 6

Operation: `print(getHexRep(98760, 4))`

And the program should return you: ValueError: not enough digits

Please be aware, the teaching assistants will use “import” command to import your program and execute some testing code to examine your program. **The name of your functions, class object, including the name of corresponding methods, should be exactly as same as the name we shown here.** Otherwise it will pop out an error and terminate the testing procedure. In that case, your homework will be considered to be failed. **Please double check your program carefully.**

Hand in procedure:

As we had mentioned in the lecture, you should list all of your collaborators in your programs. Here is the template:

```

6
7 # =====
8 #
9 # Homework assignment #0
10 # Student: Your name, 你的名字
11 # Student ID: Your student ID, 你的學號
12 # Collaborators: John Doe, Jane Doe, ...
13 #
14 # =====
15 # Your code goes here...
16
17 |

```

Please save your code as a “.py” file, where the file name should follow this format:

108A_**hw#?**_**YourID**.py

For example,

108A_**hw#1**_**0180506**.py

Please be aware. **We are not going to accept any homework file with wrong file name or without signature.** Please double check the content of your file.

Once you have accomplished your works, you can upload your homework to the “New e3” system. There will be a section for uploading your homework.